Precise Orbit Determination of GNSS Satellites — Algorithm Description

Notation

Symbol	Meaning
s, s_i	satellite index, $i = 1 \dots n$
b, b_j, b_R	ground reference station index, $j = 1 \dots m$; b_R is the reference clock
t_k, t_{k+1}	filter epochs
t_{k+1}^s	signal transmit time of satellite *s* for the epoch t_{k+1}
$\mathbf{r}$, $\mathbf{v}$, $\mathbf{a}$	position, velocity, acceleration
ECI, ECEF	Earth-centred inertial / Earth-centred Earth-fixed frame
$\boldsymbol{\kappa}$	orbit-dynamics model parameters (SRP, empirical forces, ...)
$\boldsymbol{\Phi}$	state transition matrix
ρ, φ	pseudorange, carrier phase (ionosphere-free combination)
T, Z^0, M	tropospheric delay, zenith tropospheric delay, mapping function
B	ionosphere-free combination ambiguity
dt_s, dt_b	satellite clock offset, receiver clock offset
c, ω_e, GM_E	speed of light, Earth rotation rate, Earth gravitational constant

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1. Satellite state and equation of motion

The estimated state of satellite *s* is

$$\mathbf{x}_s = \begin{bmatrix} \mathbf{r}_s^{\mathrm{ECI}} \\ \mathbf{v}_s^{\mathrm{ECI}} \\ \boldsymbol{\kappa}_s \end{bmatrix} \tag{1}$$

where $\mathbf{r}_s^{\mathrm{ECI}}$ and $\mathbf{v}_s^{\mathrm{ECI}}$ are the position and velocity of satellite *s* in the ECI frame, and $\boldsymbol{\kappa}_s$ denotes the orbit-dynamics model parameters of satellite *s* (generally solar-radiation-pressure parameters, empirical force model parameters, and so on).

The first-order differential equation of the estimated parameters $\mathbf{x}_s$ of satellite *s* with respect to time t is

$$\dot{\mathbf{x}}_s = \begin{bmatrix} \dot{\mathbf{r}}_s^{\mathrm{ECI}} \\ \dot{\mathbf{v}}_s^{\mathrm{ECI}} \\ \dot{\boldsymbol{\kappa}}_s \end{bmatrix} = \begin{bmatrix} \mathbf{v}_s^{\mathrm{ECI}} \\ \mathbf{a}_s^{\mathrm{ECI}} \\ \mathbf{0} \end{bmatrix} \tag{2}$$

where $\mathbf{a}_s^{\mathrm{ECI}}$ is the acceleration of satellite *s*, and the orbit-dynamics model parameters $\dot{\boldsymbol{\kappa}}_s$ are treated as random constants,

$$\mathbf{a}_s^{\mathrm{ECI}} = -\frac{GM_E}{r_s^3}\,\mathbf{r}_s + \mathbf{a}_{geo} + \mathbf{a}_{3rdbody} + \mathbf{a}_{srp} + \mathbf{a}_{rel} + \mathbf{a}_{amt} + \boldsymbol{\epsilon} \tag{3}$$

- r_s: the distance between satellite *s* and the Earth's centre of mass
- GM_E: the Earth's gravitational constant
- $\mathbf{a}_{geo}$: Earth gravity correction (Earth gravity field models such as EGM08)
- $\mathbf{a}_{3rdbody}$: third-body attraction
- $\mathbf{a}_{srp}$: solar radiation pressure
- $\mathbf{a}_{rel}$: relativistic effect
- $\mathbf{a}_{amt}$: atmospheric drag
- $\boldsymbol{\epsilon}$: other unmodelled acceleration

Assume that at time t_k the state parameters of satellite *s* are $\mathbf{x}_s(t_k)$; then at time t_{k+1} the state parameters of satellite *s* are

$$\mathbf{x}_s(t_{k+1}) = \int_{t_k}^{t_{k+1}} \dot{\mathbf{x}}_s\, dt \tag{4}$$

The equation above is solved by numerical integration. The commonly used numerical integration methods are

- Runge-Kutta methods of various orders (RK4, RK5, RK8, ...)
- Adams-Bashforth / Adams-Moulton predictor-corrector methods
- the Bulirsch-Stoer method
- variable-order variable-step methods (VSVO56, ...)
- symplectic integration methods

Suppose that the transition matrix of the state parameters of satellite *s* from t_k to t_{k+1} is $\boldsymbol{\Phi}_s(t_k, t_{k+1})$; when the integration interval is small, we have

$$\boldsymbol{\Phi}_s(t_k,\, t_{k+1}) = \frac{\partial \mathbf{x}_s(t_{k+1})}{\partial \mathbf{x}_s(t_k)} \tag{5}$$

At this point $\partial \mathbf{x}_s(t_{k+1}) / \partial \mathbf{x}_s(t_k)$ can be obtained by numerical differentiation. $\boldsymbol{\Phi}_s(t_k, t_{k+1})$ can also be computed by integrating the following variational equations, that is,

$$\frac{d\boldsymbol{\Phi}_s(t_k, t)}{dt} = \mathbf{A}\,\boldsymbol{\Phi}_s(t_k, t) \tag{6}$$

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \dfrac{\partial \mathbf{a}_s^{\mathrm{ECI}}(t)}{\partial \mathbf{r}_s^{\mathrm{ECI}}(t)} & \dfrac{\partial \mathbf{a}_s^{\mathrm{ECI}}(t)}{\partial \mathbf{v}_s^{\mathrm{ECI}}(t)} & \dfrac{\partial \mathbf{a}_s^{\mathrm{ECI}}(t)}{\partial \boldsymbol{\kappa}(t)} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{7}$$

$$\boldsymbol{\Phi}_s(t_k,\, t_{k+1}) = \int_{t_k}^{t_{k+1}} \mathbf{A}\, \boldsymbol{\Phi}_s(t_k, t)\, dt \tag{8}$$

The numerical integration method is the same as in eq. (4).

2. Kalman filter estimation of the satellite state

At time t_{k+1} the ground reference station *b* observes satellite *s*; the observations are the ionosphere-free combination of pseudorange and carrier phase, that is, $\rho_b^s(t_{k+1})$ and $\varphi_b^s(t_{k+1})$. The ECEF coordinates of ground reference station *b* are $\mathbf{r}_{b,\mathrm{ECEF}}$. The ECEF coordinates of station *b* are first transformed into the ECI frame, that is,

$$\mathbf{r}_{b,\mathrm{ECI}}(t_{k+1}) = \mathbf{U}(t_{k+1}) \cdot \mathbf{R}_z(-dt_b \cdot \omega_e) \cdot \mathbf{r}_{b,\mathrm{ECEF}} \tag{9}$$

where $\mathbf{U}(t_{k+1})$ is the ECEF-to-ECI transformation matrix at t_{k+1}, ω_e is the Earth rotation rate and dt_b is the receiver clock offset of ground reference station *b*.

The satellite signal propagates to the receiver of ground reference station *b* at time t_{k+1}; the signal transmit time of satellite *s* is therefore

$$t_{k+1}^s = t_{k+1} - \left(\frac{\rho_b^s(t_{k+1})}{c} + dt_s \right) \tag{10}$$

where dt_s is the satellite clock offset and c is the speed of light. At t_{k+1}^s the state parameters of satellite *s* are

$$\mathbf{x}_s\left(t_{k+1}^s\right) = \int_{t_{k+1}^s}^{t_{k+1}} \begin{bmatrix} \mathbf{v}_s^{\mathrm{ECI}}(t) \\ \mathbf{a}_s^{\mathrm{ECI}}(t) \\ \mathbf{0} \end{bmatrix} dt \tag{11}$$

Since the propagation time of the satellite signal is generally about 70 ms, the integration here runs backwards from t_{k+1} to t_{k+1}^s, which reduces the computational load.

$$\rho_b^s(t_{k+1}) = \left\| \mathbf{r}_{b,\mathrm{ECI}}(t_{k+1}) - \mathbf{x}_s(t_{k+1}^s) \right\| + T_b^s(t_{k+1}) + c \cdot (dt_b - dt_s) + \varepsilon_\rho \tag{12}$$

$$\varphi_b^s(t_{k+1}) = \left\| \mathbf{r}_{b,\mathrm{ECI}}(t_{k+1}) - \mathbf{x}_s(t_{k+1}^s) \right\| + T_b^s(t_{k+1}) + c \cdot (dt_b - dt_s) + B_b^s(t_{k+1}) + \varepsilon_\varphi \tag{13}$$

where $T_b^s(t_{k+1})$ is the tropospheric delay, $B_b^s(t_{k+1})$ is the ionosphere-free combination ambiguity, and ε_ρ and ε_φ are the pseudorange and carrier-phase observation errors. $T_b^s(t_{k+1})$ can be written as the product of the zenith tropospheric delay and a mapping function; the zenith tropospheric delay $Z_b^0(t_{k+1})$ is taken as an estimated parameter, and the coordinates of the ground reference station $\mathbf{r}_{b,\mathrm{ECI}}(t_{k+1})$ are generally known (provided by IGS weekly solutions).

Assume that at time t_{k+1} the satellites to be estimated are $s_1, s_2, s_3, \ldots, s_n$ and the ground reference stations are $b_1, b_2, b_3, \ldots, b_m$; the Kalman filter parameters to be estimated are then

$$\begin{aligned} \mathbf{X} = \big(& \mathbf{x}_{s_1}^T, \ldots, \mathbf{x}_{s_n}^T;\;\; dt_{s_1}, \ldots, dt_{s_n};\;\; dt_{b_1}, \ldots, dt_{b_m}; \\ & B_{b_1}^{s_1}, B_{b_2}^{s_1}, \ldots, B_{b_m}^{s_1};\;\; B_{b_1}^{s_2}, \ldots, B_{b_m}^{s_2}; \\ & \ldots;\;\; Z_{b_1}^0, Z_{b_2}^0, \ldots, Z_{b_m}^0 \big)^T \end{aligned} \tag{14}$$

The original prints $\mathbf{X}$ as a single tall column; it is shown here as the transpose of a row so that the equation fits the page width.

The Jacobian matrices of the pseudorange and carrier phase $\rho_{b_j}^{s_i}(t_{k+1})$ and $\varphi_{b_j}^{s_i}(t_{k+1})$ with respect to the parameters to be estimated are

$$\begin{aligned}\mathbf{H}_{\rho,b_j}^{s_i} = \Big[\cdots \ & \frac{\partial \rho_{b_j}^{s_i}(t_{k+1})}{\partial \mathbf{x}_{s_i}(t_{k+1}^{s_i})} \cdot \frac{\partial \mathbf{x}_{s_i}(t_{k+1}^{s_i})}{\partial \mathbf{x}_{s_i}(t_{k+1})} \ \cdots \\ & \cdots \ \frac{\partial \rho_{b_j}^{s_i}(t_{k+1})}{\partial dt_{s_i}} \ \cdots \ \frac{\partial \rho_{b_j}^{s_i}(t_{k+1})}{\partial dt_{b_j}} \ \cdots \\ & \cdots \ \frac{\partial \rho_{b_j}^{s_i}(t_{k+1})}{\partial B_{b_j}^{s_i}} \ \cdots \ M_{b_j}^{s_i} \Big]\end{aligned} \tag{15}$$

$$\begin{aligned}\mathbf{H}_{\varphi,b_j}^{s_i} = \Big[\cdots \ & \frac{\partial \varphi_{b_j}^{s_i}(t_{k+1})}{\partial \mathbf{x}_{s_i}(t_{k+1}^{s_i})} \cdot \frac{\partial \mathbf{x}_{s_i}(t_{k+1}^{s_i})}{\partial \mathbf{x}_{s_i}(t_{k+1})} \ \cdots \\ & \cdots \ \frac{\partial \varphi_{b_j}^{s_i}(t_{k+1})}{\partial dt_{s_i}} \ \cdots \ \frac{\partial \varphi_{b_j}^{s_i}(t_{k+1})}{\partial dt_{b_j}} \ \cdots \\ & \cdots \ \frac{\partial \varphi_{b_j}^{s_i}(t_{k+1})}{\partial B_{b_j}^{s_i}} \ \cdots \ M_{b_j}^{s_i} \Big]\end{aligned} \tag{16}$$

$$\frac{\partial \rho_{b_j}^{s_i}(t_{k+1})}{\partial \mathbf{x}_{s_i}(t_{k+1}^{s_i})} = \begin{bmatrix} \dfrac{x_{s_i}(t_{k+1}^{s_i})}{\left\| \mathbf{r}_{b,\mathrm{ECI}}(t_{k+1}) - \mathbf{x}_{s_i}(t_{k+1}^{s_i}) \right\|} \\ \dfrac{y_{s_i}(t_{k+1}^{s_i})}{\left\| \mathbf{r}_{b,\mathrm{ECI}}(t_{k+1}) - \mathbf{x}_{s_i}(t_{k+1}^{s_i}) \right\|} \\ \dfrac{z_{s_i}(t_{k+1}^{s_i})}{\left\| \mathbf{r}_{b,\mathrm{ECI}}(t_{k+1}) - \mathbf{x}_{s_i}(t_{k+1}^{s_i}) \right\|} \\ \mathbf{0}_{3\times 1} \\ \mathbf{0}_{k\times 1} \end{bmatrix} \tag{17}$$

$$\frac{\partial \mathbf{x}_{s_i}(t_{k+1}^{s_i})}{\partial \mathbf{x}_{s_i}(t_{k+1})} = \mathbf{\Phi}_{s_i}\left(t_{k+1},\, t_{k+1}^{s_i}\right) \tag{18}$$

$$\frac{\partial \rho_{b_j}^{s_i}(t_{k+1})}{\partial B_{b_j}^{s_i}} = 0, \qquad \frac{\partial \varphi_{b_j}^{s_i}(t_{k+1})}{\partial B_{b_j}^{s_i}} = 1 \tag{19}$$

where $M_{b_j}^{s_i}$ is the tropospheric mapping function value.

3. Time update and covariance propagation

The receiver clock offset of the ground station and the satellite clock offset are modelled as white noise or a first-order Markov process, the ionosphere-free combination ambiguity is modelled as a random constant, and the zenith tropospheric delay is modelled as a first-order Markov process. Then, from t_k to t_{k+1}, the transition matrix of the Kalman filter parameters $\mathbf{X}$ is

$$\mathbf{F}(t_k, t_{k+1}) = \mathrm{diag}(\mathbf{\Phi}_{s_1}(t_k, t_{k+1}),\, \mathbf{\Phi}_{s_2}(t_k, t_{k+1}),\ \ \mathbf{\Phi}_{s_3}(t_k, t_{k+1}),\, \ldots,\, \mathbf{\Phi}_{s_n}(t_k, t_{k+1}),\, 1, 1, \ldots, 1) \tag{20}$$

where $\mathrm{diag}(*)$ denotes the formation of a diagonal matrix.

At time t_{k+1} the covariance matrix of the parameters to be estimated $\mathbf{X}$ is

$$\mathbf{P}(t_{k+1}) = \mathbf{F}(t_k, t_{k+1}) \cdot \mathbf{P}(t_k) \cdot \mathbf{F}^T(t_k, t_{k+1}) + \mathbf{Q}(t_{k+1}) \tag{21}$$

where $\mathbf{Q}(t_{k+1})$ is the system noise matrix.

If the ground reference station b_R is set as the reference clock, then

$$\frac{\partial \rho_{b_R}^{s_i}(t_{k+1})}{\partial dt_{b_R}} = 0 \tag{22}$$

$$\frac{\partial \varphi_{b_R}^{s_i}(t_{k+1})}{\partial dt_{b_R}} = 0 \tag{23}$$